SUKRUTHA M

Embedded Systems Engineer

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CAREER OBJECTIVE

Enthusiastic Embedded Systems fresher with hands-on experience in microcontroller programming, hardware interfacing, and communication protocols. Proficient in Embedded C and firmware development, with a strong interest in embedded system design. Eager to contribute technical skills toward developing efficient and reliable embedded solutions.

TECHNICAL SKILLS

- **Programming:** C, Embedded C, Python
- **Microcontrollers:** STM32, MSP430, AVR ATmega32, PIC16F, LPC2148, Arduino Uno, Raspberry Pi Pico
- **Development Tools:** STM32CubeIDE, Keil µVision, MPLAB X IDE, Code Composer Studio, Arduino IDE
- **Communication Protocols:** UART, SPI, I2C
- **Embedded Skills:** GPIO, ADC, PWM, Motor Driver Interfacing, Breadboard Prototyping
- **PCB Design:** KiCad, Altium Designer
- **Embedded Linux:** Raspberry Pi GPIO, Basic Shell Scripting

EXPERIENCE

Hardware Intern

Webyfy IoT Pvt. Ltd., Trivandrum, Kerala | August 2026 – October 2026

- Simulated voltage and current sensor circuits using LTspice.
- Verified circuit outputs and analyzed simulation waveforms and graphs.
- Worked with ESP32 for Ethernet communication.
- Learned Wi-Fi connectivity, MQTT messaging, and HTTP communication for IoT applications.
- Gained practical exposure to electronic circuit simulation, testing, and embedded hardware.

PROJECTS

1. Realization of a PID Algorithm for DC Motor Speed Control

- Developed a closed-loop PID control system using Arduino to regulate DC motor speed.
- Implemented real-time error correction between desired and actual speed.
- Tuned Kp, Ki, and Kd parameters to reduce steady-state error and overshoot.
- Analyzed performance using speed vs. time graphs and error metrics.

2. Self-Balancing Boat Stabilization System

- Developed a real-time boat stabilization system using ESP32, MPU6050, and a stepper motor.
- Implemented FreeRTOS for multitasking and interfaced the MPU6050 using the I2C protocol.
- Controlled the stepper motor based on the measured roll angle to automatically stabilize the boat.

CERTIFICATION

Certified Embedded Systems Engineer

IPCS Global, Trivandrum | 2025 – Present

- Completed structured hands-on training in Embedded C programming and microcontroller architecture.
- Worked with GPIO, ADC, PWM, and peripheral interfacing for embedded control applications.
- Hands-on experience with Arduino, AVR, PIC, ARM, MSP430, STM32, Raspberry Pi Pico, and Raspberry Pi platforms.
- Developed and tested embedded applications using microcontroller and microprocessor-based systems.
- Familiar with electronic schematics, basic circuit design, PCB fundamentals, and board layout concepts.

EDUCATION

- Bachelor of Science (BSc) in Instrumentation – NSS College, Nemmara, Kerala | 2022–2025
- 12th | GHSS Peringottukurissi, Kerala State Board | 2022
- 10th | HSS Kuthanur, Kerala State Board | 2020

LANGUAGES

- English, Malayalam